

If $w = F(x, y, z)$ has continuous partial derivatives of any order, then analogous to (3), the mixed partial derivatives are equal:

$$F_{xyz} = F_{yzx} = F_{zyx}, \quad F_{xxy} = F_{yxx} = F_{xyx},$$

and so on.

9.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–6, sketch some of the level curves associated with the given function.

1. $f(x, y) = x + 2y$
2. $f(x, y) = y^2 - x$
3. $f(x, y) = \sqrt{x^2 - y^2 - 1}$
4. $f(x, y) = \sqrt{36 - 4x^2 - 9y^2}$
5. $f(x, y) = e^{y-x^2}$
6. $f(x, y) = \tan^{-1}(y - x)$

In Problems 7–10, describe the level surfaces but do not graph.

7. $F(x, y, z) = \frac{x^2}{9} + \frac{z^2}{4}$
8. $F(x, y, z) = x^2 + y^2 + z^2$
9. $F(x, y, z) = x^2 + 3y^2 + 6z^2$
10. $F(x, y, z) = 4y - 2z + 1$
11. Graph some of the level surfaces associated with $F(x, y, z) = x^2 + y^2 - z^2$ for $c = 0$, $c > 0$, and $c < 0$.
12. Given that

$$F(x, y, z) = \frac{x^2}{16} + \frac{y^2}{4} + \frac{z^2}{9},$$

find the x -, y -, and z -intercepts of the level surface that passes through $(-4, 2, -3)$.

In Problems 13–32, find the first partial derivatives of the given function.

13. $z = x^2 - xy^2 + 4y^5$

14. $z = -x^3 + 6x^2y^3 + 5y^2$

15. $z = 5x^4y^3 - x^2y^6 + 6x^5 - 4y$

16. $z = \tan(x^3y^2)$

17. $z = \frac{4\sqrt{x}}{3y^2 + 1}$

18. $z = 4x^3 - 5x^2 + 8x$

19. $z = (x^3 - y^2)^{-1}$

20. $z = (-x^4 + 7y^2 + 3y)^6$

21. $z = \cos^2 5x + \sin^2 5y$

22. $z = e^{x^2 \tan^{-1} y^2}$

23. $f(x, y) = xe^{x^3y}$

24. $f(\theta, \phi) = \phi^2 \sin \frac{\theta}{\phi}$

25. $f(x, y) = \frac{3x - y}{x + 2y}$

26. $f(x, y) = \frac{xy}{(x^2 - y^2)^2}$

27. $g(u, v) = \ln(4u^2 + 5v^3)$

28. $h(r, s) = \frac{\sqrt{r}}{s} - \frac{\sqrt{s}}{r}$

29. $w = 2\sqrt{xy} - ye^{y/z}$

30. $w = xy \ln(xz)$

31. $F(u, v, x, t) = u^2w^2 - uv^3 + vw \cos(ut^2) + (2x^2t)^4$

32. $G(p, q, r, s) = (p^2q^3)r^4s^5$

In Problems 33 and 34, verify that the given function satisfies Laplace's equation:

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

33. $z = \ln(x^2 + y^2)$

34. $z = e^{x^2 - y^2} \cos 2xy$

In Problems 35 and 36 verify that the given function satisfies the **wave equation**:

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}.$$

35. $u = \cos at \sin x$

36. $u = \cos(x + at) + \sin(x - at)$

37. The molecular concentration $C(x, t)$ of a liquid is given by $C(x, t) = t^{-1/2} e^{-x^2/4t}$. Verify that this function satisfies the **diffusion equation**:

$$\frac{k}{4} \frac{\partial^2 C}{\partial x^2} = \frac{\partial C}{\partial t}.$$

38. The pressure P exerted by an enclosed ideal gas is given by $P = k(T/V)$, where k is a constant, T is temperature, and V is volume. Find:

(a) the rate of change of P with respect to V ,

(b) the rate of change of V with respect to T , and

(c) the rate of change of T with respect to P .

In Problems 39–48, use the Chain Rule to find the indicated partial derivatives.

39. $z = e^{uv^2}; u = x^3, v = x - y^2; \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

40. $z = u^2 \cos 4v; u = x^2 y^3, v = x^3 + y^3; \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

41. $z = 4x - 5y^2; x = u^4 - 8v^3, y = (2u - v)^2; \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}$

42. $z = \frac{x-y}{x+y}; x = \frac{u}{v}, y = \frac{v^2}{u}; \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}$
43. $w = (u^2 + v^2)^{3/2}; u = e^{-t} \sin \theta, v = e^{-t} \cos \theta; \frac{\partial w}{\partial t}, \frac{\partial w}{\partial \theta}$
44. $w = \tan^{-1} \sqrt{uv}; u = r^2 - s^2, v = r^2 s^2; \frac{\partial w}{\partial r}, \frac{\partial w}{\partial s}$
45. $R = rs^2 t^4; r = ue^{v^2}, s = ve^{-u^2}, t = e^{u^2 v^2}; \frac{\partial R}{\partial u}, \frac{\partial R}{\partial v}$
46. $Q = \ln(pqr); p = t^2 \sin^{-1} x, q = \frac{x}{t^2}, r = \tan^{-1} \frac{x}{t}; \frac{\partial Q}{\partial x}, \frac{\partial Q}{\partial t}$
47. $w = \sqrt{x^2 + y^2}; x = \ln(rs + tu),$
 $y = \frac{t}{u} \cosh rs; \frac{\partial w}{\partial t}, \frac{\partial w}{\partial r}, \frac{\partial w}{\partial u}$
 $s = p^2 + q^2 - r^2 + 4t; p = \phi e^{3\theta}, q = \cos(\phi + \theta), r = \phi \theta^2,$
48. $t = 2\phi + 8\theta; \frac{\partial s}{\partial \phi}, \frac{\partial s}{\partial \theta}$

In Problems 49–52, use (8) to find the indicated derivative.

49. $z = \ln(u^2 + v^2); u = t^2, v = t^{-2}; \frac{dz}{dt}$
50. $z = u^3 v - uv^4; u = e^{-5t}, v = \sec 5t; \frac{dz}{dt}$
51. $w = \cos(3u + 4v); u = 2t + \frac{\pi}{2}, v = -t - \frac{\pi}{4}; \frac{dw}{dt} \Big|_{t=\pi}$
52. $w = e^{xy}; x = \frac{4}{2t+1}, y = 3t+5; \frac{dw}{dt} \Big|_{t=0}$
53. If $u = f(x, y)$ and $x = r \cos \theta, y = r \sin \theta$, show that **Laplace's equation** $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$ becomes

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

54. Van der Waals' equation of state for the real gas CO₂ is

$$P = \frac{0.08T}{V - 0.0427} - \frac{3.6}{V^2}.$$

If dT/dt and dV/dt are rates at which the temperature and volume change, respectively, use the Chain Rule to find dP/dt .

55. The equation of state for a thermodynamic system is $F(P, V, T) = 0$, where P , V , and T are pressure, volume, and temperature, respectively. If the equation defines V as a function of P and T , and also defines T as a function of V and P , show that

$$\frac{\partial V}{\partial T} = -\frac{\frac{\partial F}{\partial T}}{\frac{\partial F}{\partial V}} = -\frac{1}{\frac{\partial T}{\partial V}}.$$

56. The voltage across a conductor is increasing at a rate of 2 volts/min and the resistance is decreasing at a rate of 1 ohm/min. Use $I = E/R$ and the Chain Rule to find the rate at which the current passing through the conductor is changing when $R = 50$ ohms and $E = 60$ volts.
57. The length of the side labeled x of the triangle in **FIGURE 9.4.6** increases at a rate of 0.3 cm/s, the side labeled y increases at a rate of 0.5 cm/s, and the included angle θ increases at a rate of 0.1 rad/s. Use the Chain Rule to find the rate at which the area of the triangle is changing at the instant $x = 10$ cm, $y = 8$ cm, and $\theta = \pi/6$.

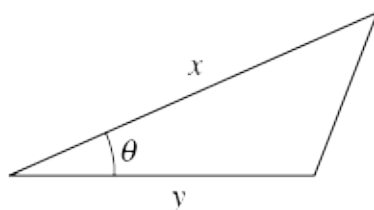


FIGURE 9.4.6 Triangle in Problem 57

58. A particle moves in 3-space so that its coordinates at any time are $x = 4 \cos t$, $y = 4 \sin t$, $z = 5t$, $t \geq 0$. Use the Chain Rule to

find the rate at which its distance

$$w = \sqrt{x^2 + y^2 + z^2}$$

from the origin is changing at $t = 5\pi/2$ seconds.